Linear Algebra and Its Applications

Section 7.1: Diagonalization of Symmetric Matrices

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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7.1 Diagonalization of symmetric matrices

A real matrix is symmetric when $A^T = A$, and an orthogonal matrix satisfies $P^T P = I$, so $P^{-1} = P^T$. These are different definitions. The spectral theorem says that a real square matrix has an orthonormal eigenvector basis exactly when it is symmetric. To orthogonally diagonalize a symmetric matrix:

1. Find its real eigenvalues, with multiplicities.
2. For each λ, solve $(A - \lambda I)v = 0$ by row reduction.
3. Within a repeated eigenspace, use Gram–Schmidt if needed. Eigenvectors belonging to distinct eigenvalues are already perpendicular.
4. Normalize each nonzero vector v_i to $u_i = v_i/\|v_i\|$. Set $P = [u_1 \ \cdots \ u_n]$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ in the same order. Then $A = PDP^T$.

In the computations below, the columns of V are unnormalized orthogonal eigenvectors. A diagonal factor rescales the columns to form P. Every displayed diagonalization was checked exactly by $AP = PD$ and $P^T P = I$. Changing a column's sign or reordering columns together with D gives another equally valid answer.

Practice problem 7.1.P1: A square of a symmetric matrix is symmetric

Using the reverse-order transpose rule, $(A^2)^T = (AA)^T = A^T A^T = AA = A^2$, since $A^T = A$. Thus A^2 is symmetric. The reversal creates no difficulty here because the two factors are the same matrix.

Practice problem 7.1.P2: Square an orthogonal diagonalization

If $A = PDP^T$ with $P^T P = I$, then $A^2 = PDP^T PDP^T = PD^2 P^T$. The same P is orthogonal and D^2 is diagonal, so this is an orthogonal diagonalization of A^2.

Exercise 7.1.1: Match entries across the diagonal

The two off-diagonal entries are both 5, so $\begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix}^T$ is the original matrix. It is symmetric.

Exercise 7.1.2: Opposite signs do not match

The upper-right entry is 5 and the lower-left entry is -5. They would have to be equal for symmetry. Therefore the matrix is not symmetric.

Exercise 7.1.3: One unequal pair is enough

Here $a_{12} = 2$ but $a_{21} = 4$. Thus $A^T \neq A$, so the matrix is not symmetric, despite the repeated entries within each row.

Exercise 7.1.4: Check all three off-diagonal pairs

The paired entries are $a_{12} = a_{21} = 8$, $a_{13} = a_{31} = 3$, and $a_{23} = a_{32} = -2$. The diagonal is unchanged by transpose. Thus this matrix is symmetric.

Exercise 7.1.5: An upper triangular matrix with nonzero off-diagonal entries

The entry $a_{12} = 2$ has counterpart $a_{21} = 0$. Thus the matrix is not symmetric. An upper triangular symmetric matrix would have to be diagonal, because its entries below and above the diagonal must both vanish.

Exercise 7.1.6: Symmetry requires a square matrix

The given array has three rows and four columns. Its transpose has four rows and three columns, so the two cannot be equal. It is not symmetric. The repeated numerical pattern does not overcome the shape mismatch.

Exercise 7.1.7: Orthonormal decimal columns

The two columns have squared lengths $.6^2 + .8^2 = 1$ and $.8^2 + (-.6)^2 = 1$. Their dot product is $.6(.8) + .8(-.6) = 0$. Thus A is orthogonal, and

$$A^{-1} = A^T = \begin{bmatrix} .6 & .8 \\ .8 & -.6 \end{bmatrix}.$$

Here the matrix happens to be symmetric too, so it is its own inverse.

Exercise 7.1.8: A rotation has its transpose as inverse

Each column has squared length $1/2 + 1/2 = 1$, and the dot product is $-1/2 + 1/2 = 0$. Hence the matrix is orthogonal and

$$A^{-1} = A^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Geometrically this reverses the 45° rotation of the original matrix.

Exercise 7.1.9: Perpendicular but not unit columns

The columns $(-5, 2)^T$ and $(2, 5)^T$ are perpendicular, but each has squared length $25 + 4 = 29$. Thus $A^T A = 29I$, not I. The matrix is not orthogonal. Having an inverse is a separate issue; orthogonality specifically requires the columns to be normalized.

Exercise 7.1.10: Another missing normalization

Each column has squared length $1 + 4 + 4 = 9$, and distinct columns have dot product $-2 - 2 + 4 = 0$. Therefore $A^T A = 9I$, not I. The given matrix is not orthogonal; dividing it by 3 would make it so.

Exercise 7.1.11: Verify a less obvious orthogonal matrix

Let the columns be c_1, c_2, c_3. Their squared lengths are

$$\|c_1\|^2 = 4/9 + 5/9 = 1, \quad \|c_2\|^2 = 4/9 + 1/5 + 16/45 = 1, \quad \|c_3\|^2 = 1/9 + 4/5 + 4/45 = 1.$$

Their pairwise dot products are $4/9 - 4/9 = 0$, $2/9 - 2/9 = 0$, and $2/9 - 2/5 + 8/45 = 0$. Thus all columns are orthonormal and the inverse is the transpose:

$$A^{-1} = \begin{bmatrix} 2/3 & 0 & \sqrt{5}/3 \\ 2/3 & 1/\sqrt{5} & -4/\sqrt{45} \\ 1/3 & -2/\sqrt{5} & -2/\sqrt{45} \end{bmatrix}.$$

Exercise 7.1.12: An orthogonal signed matrix

Write $A = H/2$, where H has the printed entries 1 or -1. Each column of H has squared length 4, and every pair of different columns has two agreeing signs and two disagreeing signs, so dot product zero. Thus $A^T A = I$, and

$$A^{-1} = A^T = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}.$$

Exercise 7.1.13: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t-4)(t-2)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2$. $\lambda = 4$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (2, 4).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(2, 4).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.14: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t-6)(t+4)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -4$: row reduction gives $x_1 = -x_2$. $\lambda = 6$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-4, 6).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(-4, 6).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.15: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $t(t-17)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = \frac{x_2}{4}$. $\lambda = 17$: row reduction gives $x_1 = -4x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 4 \\ 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (0, 17).$$

Their squared lengths are $(17, 17)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{17}} V, \qquad D = \operatorname{diag}(0, 17).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.16: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t - 25)(t + 25)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -25$: row reduction gives $x_1 = -\frac{4x_2}{3}$. $\lambda = 25$: row reduction gives $x_1 = \frac{3x_2}{4}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-25, 25).$$

Their squared lengths are $(25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-25, 25).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.17: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = 0$. $\lambda = 2$: row reduction gives $x_1 = x_3, \quad x_2 = -2x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(-2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.18: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -50$: row reduction gives $x_1 = \frac{3x_2}{4}, \quad x_3 = 0$. $\lambda = 3$: row reduction gives $x_1 = 0, \quad x_2 = 0$. $\lambda = 25$: row reduction gives $x_1 = -\frac{4x_2}{3}, \quad x_3 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 3 & 0 & 4 \\ 4 & 0 & -3 \\ 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-50, 3, 25).$$

Their squared lengths are $(25, 1, 25)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{5}, 1, \frac{1}{5}), \qquad D = \operatorname{diag}(-50, 3, 25).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.19: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = -\frac{x_3}{2}$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_2}{2} + x_3$. Within the 7-eigenspace, start with $(1, 0, 1)^T$ and $(-1/2, 1, 0)^T$. Subtracting the projection of the latter onto the former gives $(-1/4, 1, 1/4)^T$, a multiple of $(-1, 4, 1)^T$. This is the Gram–Schmidt step that produces the two orthogonal columns used below.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 4 \\ -2 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 7, 7).$$

Their squared lengths are $(9, 2, 18)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{3}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}), \qquad D = \operatorname{diag}(-2, 7, 7).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.20: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 13$: row reduction gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 7, 13).$$

Their squared lengths are $(9, 9, 9)$. Hence an explicit answer is

$$P = \frac{1}{3}V, \qquad D = \operatorname{diag}(1, 7, 13).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.21: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -x_3, \quad x_2 = -x_4$. $\lambda = 5$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$. $\lambda = 9$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (1, 1, 5, 9).$$

Their squared lengths are $(2, 2, 4, 4)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{1}{2}), \qquad D = \operatorname{diag}(1, 1, 5, 9).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.22: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = 0, \quad x_2 = -x_4, \quad x_3 = 0$. $\lambda = 2$: row reduction gives $x_2 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (0, 2, 2, 2).$$

Their squared lengths are $(2, 1, 1, 2)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, 1, 1, \frac{\sqrt{2}}{2}), \qquad D = \operatorname{diag}(0, 2, 2, 2).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.23: An orthogonal eigenvector basis

The printed vector $v = (1, 1, 1)^T$ satisfies $Av = 5v$, so its eigenvalue is 5. The exercise also asks to verify 2 is an eigenvalue; the vector $(1, -1, 0)^T$ satisfies $A(1, -1, 0)^T = 2(1, -1, 0)^T$. Thus the two assertions in the wording refer to different eigenvectors.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2 - x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.24: An orthogonal eigenvector basis

Multiplication gives $Av_1 = 10v_1$ for the supplied $v_1 = (-2, 2, 1)^T$, and $Av_2 = v_2$ for $v_2 = (1, 1, 0)^T$. For the repeated eigenvalue 1, $(1, -1, 4)^T$ is a second eigenvector orthogonal to v_2.

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = x_2 + \frac{x_3}{2}$. $\lambda = 10$: row reduction gives $x_1 = -2x_3, \quad x_2 = 2x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & -2 \\ 0 & 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 1, 10).$$

Their squared lengths are $(2, 18, 9)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}, \frac{1}{3}), \qquad D = \operatorname{diag}(1, 1, 10).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.25: Symmetry, multiplicity, and a missing unit condition

(a) **True.** If $A = PDP^T$, then $A^T = (PDP^T)^T = PD^T P^T = PDP^T = A$.

(b) **True.** Symmetry gives $3u^T v = (Au)^T v = u^T Av = 4u^T v$, so $u^T v = 0$.

(c) **False.** I_n is symmetric but has only one distinct eigenvalue, namely 1 with multiplicity n.

(d) **False.** For $B = vv^T$, $B^2 = (v^T v)B$. A projection requires $B^2 = B$, which for nonzero v requires $\|v\| = 1$. For $v = 2e_1$, $B = 4e_1 e_1^T$ is not idempotent.

Exercise 7.1.26: Four consequences and nonconsequences of the spectral theorem

(a) **True.** This is the real symmetric spectral theorem.

(b) **True.** Taking the transpose of PDP^T gives the same matrix because D is real diagonal.

(c) **False.** A 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but not symmetric, so it cannot be orthogonally diagonalized over $\mathbb{R}$.

(d) **True.** In the spectral theorem the orthogonal diagonalization provides one basis vector for each occurrence of an eigenvalue on the diagonal, giving equality of geometric and algebraic multiplicities.

Exercise 7.1.27: Three symmetric products

Use the reverse-order transpose rule and the hypothesis $A^T = A$:

$$(B^T AB)^T = B^T A^T B = B^T AB, \quad (B^T B)^T = B^T B, \quad (BB^T)^T = BB^T.$$

The products have square sizes $m \times m$, $m \times m$, and $n \times n$, respectively, so these equalities indeed establish symmetry in each case.

Exercise 7.1.28: Move a symmetric matrix across a dot product

For every x, y, $(Ax) \cdot y = (Ax)^T y = x^T A^T y = x^T Ay = x \cdot (Ay)$. The step using $A^T = A$ is exactly where symmetry matters. Without it the correct expression would contain A^T on the other side.

Exercise 7.1.29: The inverse retains the eigenvectors

Write $A = PDP^T$ with P orthogonal. Since A is invertible, none of the diagonal eigenvalues in D is zero, so D^{-1} exists and is diagonal. Then $A^{-1} = PD^{-1}P^T$, an orthogonal diagonalization with the same P. Its diagonal eigenvalues are the reciprocals of the original ones.

Exercise 7.1.30: A commuting product remains symmetric

Orthogonal diagonalizability makes both A and B symmetric. Therefore $(AB)^T = B^T A^T = BA = AB$, using the additional commutation hypothesis. Thus AB is symmetric and the spectral theorem makes it orthogonally diagonalizable. Without $AB = BA$, a product of two symmetric matrices need not be symmetric.

Exercise 7.1.31: Count a repeated eigenvalue's eigenspace dimension

Let $x = Py$. Since P is invertible, $Ax = \lambda x$ is equivalent to $(D - \lambda I)y = 0$. At positions where the diagonal of D differs from λ, this forces $y_i = 0$. At its k positions equal to λ, the corresponding y_i are arbitrary. Thus the solution space has exactly k independent coordinates. Mapping by P gives the eigenspace of A and preserves its dimension, so that dimension is k. Equivalently the k corresponding columns of P form its basis.

Exercise 7.1.32: A symmetric triangular matrix is diagonal

From $A = PRP^T$, solve for $R = P^T AP$. Taking transposes gives $R^T = P^T A^T P = P^T AP = R$, so R is symmetric. But R is also upper triangular. Entries below its diagonal are zero; symmetry forces their counterpart entries above the diagonal to be zero too. Thus all off-diagonal entries vanish and R is diagonal.

Exercise 7.1.33: A spectral decomposition for Example 2

Use the orthogonal eigenvectors $a = (-1, 1, 0)^T$, $b = (-1, -1, 2)^T$, $c = (1, 1, 1)^T$, with eigenvalues $8, 6, 3$ and squared lengths $2, 6, 3$. After normalization the spectral formula gives

$$\boxed{A = 4aa^T + bb^T + cc^T}.$$

Indeed the normalization divides each outer product by its squared length; thus the three coefficients are 8/2, 6/6, 3/3. Expanding these outer products gives $\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$, the matrix from Example 2. This checks the decomposition entrywise.

Exercise 7.1.34: A spectral decomposition with a repeated eigenvalue

For Example 3, take $a = (1, 0, 1)^T$, $b = (-1, 4, 1)^T$, and $c = (-2, -1, 2)^T$. They are mutually perpendicular, with squared lengths $2, 18, 9$. The first two have eigenvalue 7, and the last has eigenvalue -2. Therefore

$$\boxed{A = \frac{7}{2}aa^T + \frac{7}{18}bb^T - \frac{2}{9}cc^T}.$$

For instance the $(1, 1)$ entry is $7/2 + 7/18 - 8/9 = 3$; expansion of all entries returns $\begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. The two separate rank-one terms for eigenvalue 7 reflect its two-dimensional eigenspace.

Exercise 7.1.35: A rank-one projection from a unit vector

(a) $Bx = (uu^T)x = u(u^T x) = (x \cdot u)u$. Since $u \cdot u = 1$, this equals $((x \cdot u)/(u \cdot u))u$, the orthogonal projection formula.

(b) $B^T = (uu^T)^T = uu^T = B$ and $B^2 = u(u^T u)u^T = uu^T = B$.

(c) $Bu = u(u^T u) = u$. The unit vector is nonzero, so it is an eigenvector with eigenvalue 1.

Exercise 7.1.36: Every symmetric idempotent matrix is an orthogonal projection

(a) Show the proposed pieces are perpendicular.

Put $\widehat{y} = By$, $z = (I - B)y$. Symmetry and idempotence give

$$z^T \widehat{y} = y^T (I - B)^T By = y^T (B - B^2)y = 0.$$

(b) Verify the stronger orthogonality needed for a projection.

Clearly $\widehat{y} \in W = \operatorname{Col} B$. For any $w \in W$, write $w = Bv$; then $z^T w = y^T (I - B)Bv = 0$. Thus $z \in W^\perp$, not merely perpendicular to one particular vector $\widehat{y}$. The identity $y = By + (y - By)$ is therefore the orthogonal decomposition into a vector in W and one in $W^\perp$. Its uniqueness proves that $By = \operatorname{proj}_W y$.

Exercise 7.1.37: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -12$: row reduction gives $x_1 = x_4, \quad x_2 = -x_4, \quad x_3 = -x_4$. $\lambda = 4$: row reduction gives $x_1 = -x_4, \quad x_2 = -x_4, \quad x_3 = x_4$. $\lambda = 10$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = 18$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-12, 4, 10, 18).$$

Their squared lengths are $(4, 4, 4, 4)$. Hence an explicit answer is

$$P = \frac{1}{2}V, \qquad D = \operatorname{diag}(-12, 4, 10, 18).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.38: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = \frac{1}{4}$: row reduction gives $x_1 = 4x_4, \quad x_2 = 2x_4, \quad x_3 = 2x_4$. $\lambda = \frac{3}{10}$: row reduction gives $x_1 = -\frac{x_4}{4}, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = \frac{x_4}{2}$. $\lambda = \frac{11}{20}$: row reduction gives $x_1 = x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = -2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 1 & 2 & 2 \\ 2 & 2 & -1 & -4 \\ 2 & -2 & -4 & 1 \\ 1 & -4 & 2 & -2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.39: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -\frac{5}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$. $\lambda = 0$: row reduction gives $x_1 = -x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = 2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = \frac{3x_3}{2} + x_4, \quad x_2 = x_3$. For the repeated eigenvalue $3/4$, the two chosen null vectors $(4, 2, 2, 1)^T$ and $(1, -2, -2, 4)^T$ have dot product $4 - 4 - 4 + 4 = 0$. Both have squared length 25, so dividing each by 5 normalizes them.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 2 & 4 & 1 \\ -4 & 1 & 2 & -2 \\ 1 & -4 & 2 & -2 \\ -2 & -2 & 1 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.40: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -28$: row reduction gives $x_1 = -\frac{x_5}{4}, \quad x_2 = -\frac{x_5}{4}, \quad x_3 = -\frac{x_5}{4}, \quad x_4 = -\frac{x_5}{4}$. $\lambda = 8$: row reduction gives $x_1 = -x_2 - x_3, \quad x_4 = 0, \quad x_5 = 0$. $\lambda = 17$: row reduction gives $x_1 = x_5, \quad x_2 = x_5, \quad x_3 = x_5, \quad x_4 = x_5$. $\lambda = 32$: row reduction gives $x_1 = -\frac{x_4}{3}, \quad x_2 = -\frac{x_4}{3}, \quad x_3 = -\frac{x_4}{3}, \quad x_5 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 \\ 1 & 0 & -2 & 1 & 1 \\ 1 & 0 & 0 & 1 & -3 \\ -4 & 0 & 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_5) = (-28, 8, 8, 17, 32).$$

Their squared lengths are $(20, 2, 6, 5, 12)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{5}}{10}, \frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{5}}{5}, \frac{\sqrt{3}}{6}), \qquad D = \operatorname{diag}(-28, 8, 8, 17, 32).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.